

FALCON's Response to Kravtel: Supply of PTL System

June 29, 2023



www.falconautotech.com

Kind Attention –

Dmitry Lednev

Offer Ref: F23-00326 ; Date 29.06.2023.

Subject – Techno-Commercial Offer for PTL system.

Dear Team,

We are pleased to submit our techno-commercial offer in response to your RFQ.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Contact Person : Sanyog Pratap Singh
Title : Asst. Manager
Mobile : +91 8750 052 591
Email : Sanyog.singh@falconautoonline.com

Company Details

Company Name : Falcon Autotech Pvt. Ltd.
Regd. Office : 57, Nimri Colony, Double Storey Flats, Delhi – 110052
Website : www.falconautotech.com
LinkedID : <https://www.linkedin.com/company/falconautotech/mycompany/>

Table of Contents

1. About Falcon	4
2. Proposed System Description.....	9
2.1 Objective	9
3. Pick/Put to Light Module	9
4. Sample Rack Layout.....	10
4.1 Type-1	10
4.2 Type-2	11
4.3 Type-3	11
5. Handheld Barcode Scanner	11
6. Falcon's WCS CONTROLIT.....	12
7. Proposed System Technical Details/BOM	13
7.1 Mechanical equipment	13
7.1.1 Setup of Type-1 (36 X 30 Lights)	13
7.1.2 Setup of Type-2 (22 X 30 Lights)	13
7.1.3 Setup of Type-3 (10 X 30 Lights)	13
7.2 Electrical Equipment	13
7.3 Control System.....	14
8. Commercial Terms.....	14
8.1 Price Sheet.....	14
9. Dispatch Timelines	14
10. Payment Terms	15
11. Warranty Period	15
12. Exclusions	16

1. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

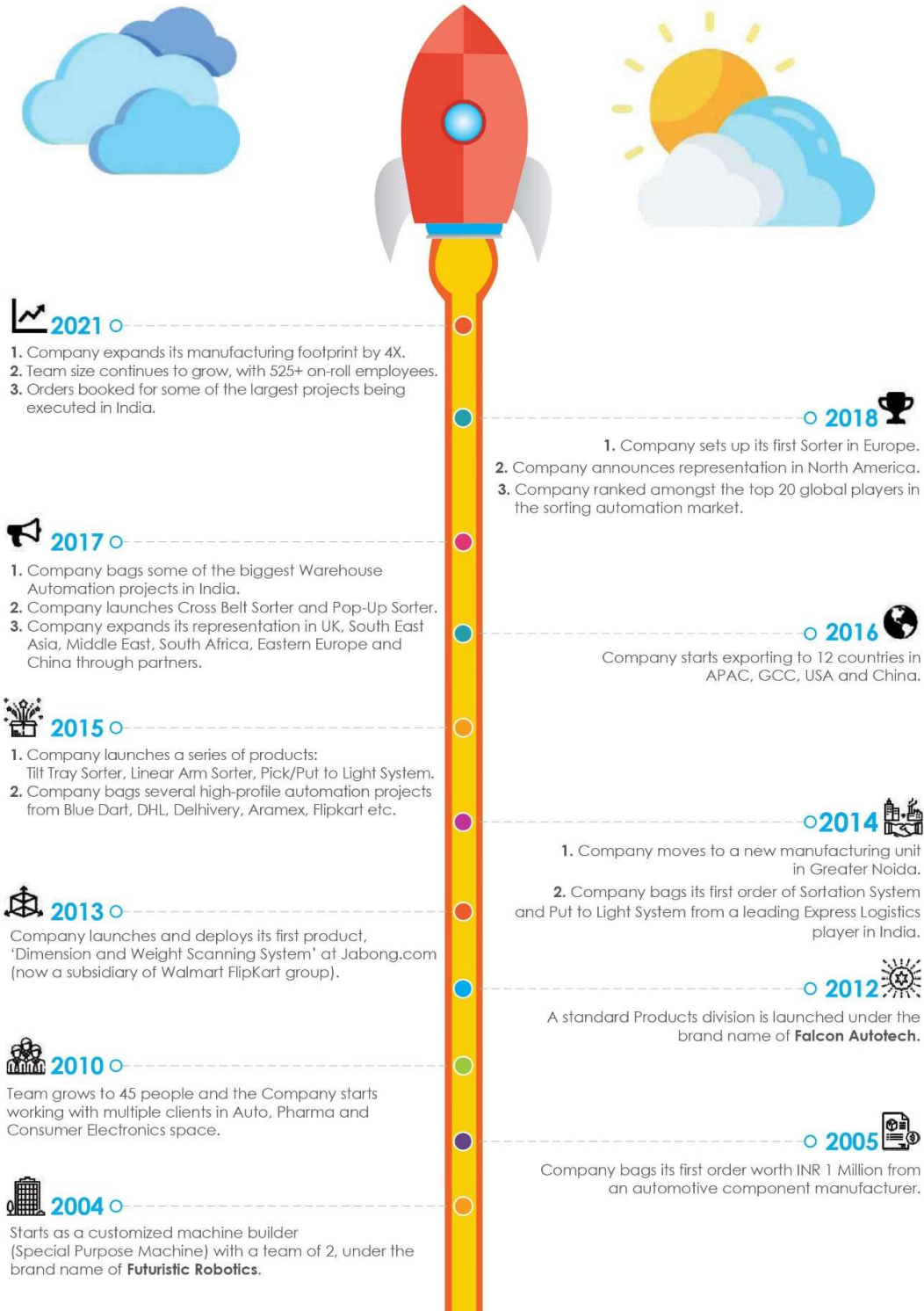
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.



COMPANY HISTORY



OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

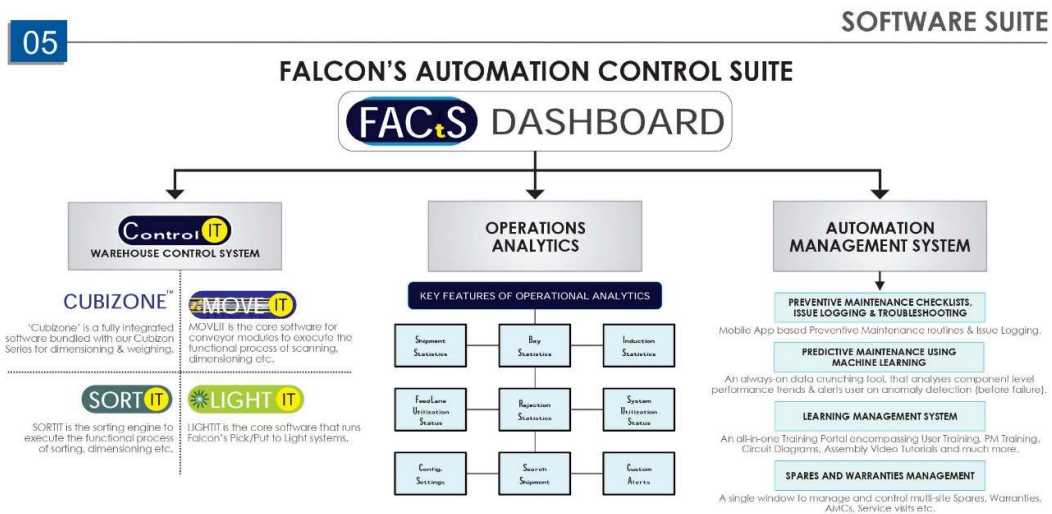
DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS



OUR PRODUCT PORTFOLIO

04				CONVEYOR AUTOMATION SOLUTIONS			
BELT CONVEYORS		ROLLER CONVEYORS		SPECIAL APPLICATION SOLUTIONS			
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve		◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit		◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor			
							



Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



Deep Learning Image Classification

DLC is an artificial Intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.

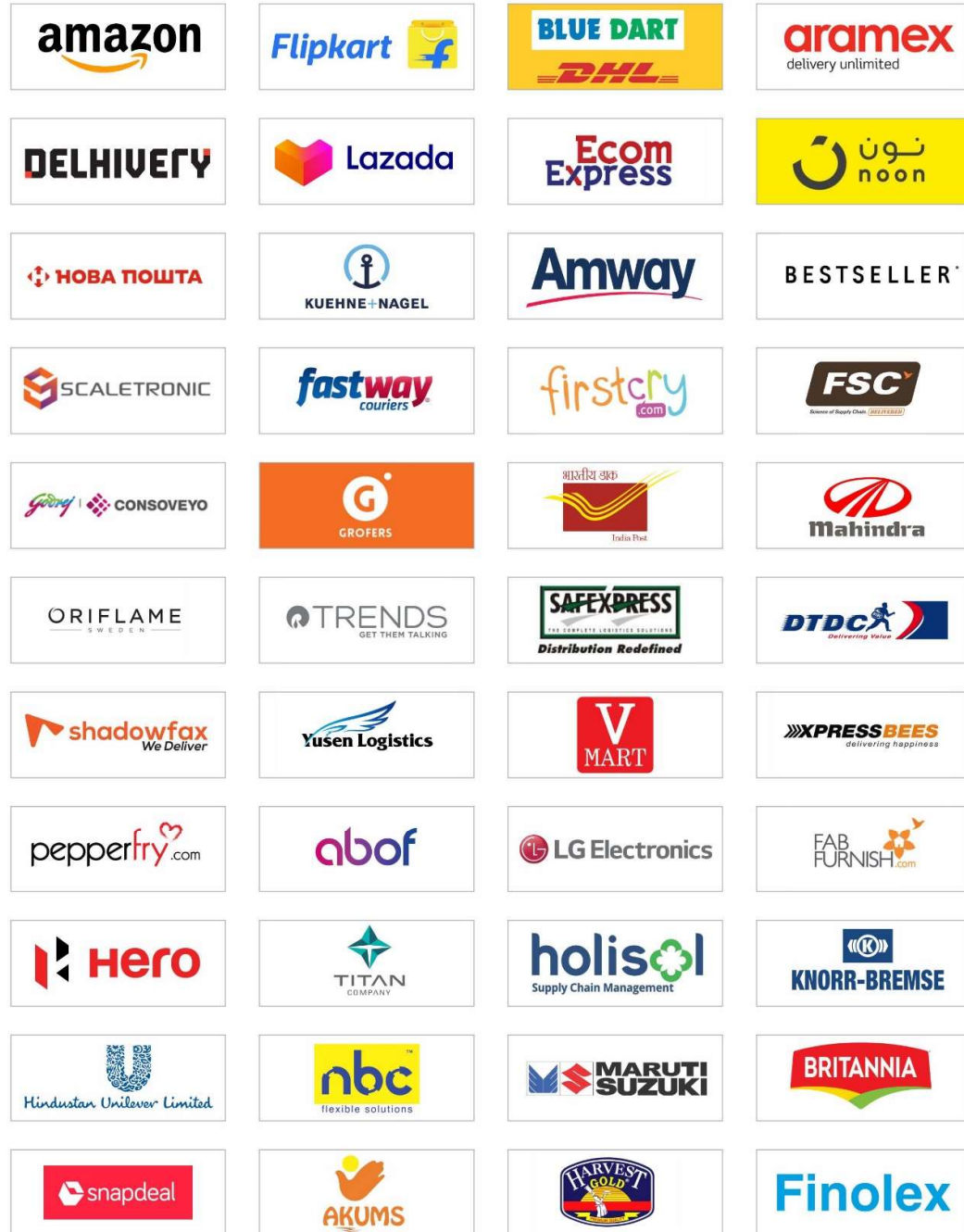


Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



2. Proposed System Description

2.1 Objective

The purpose of this proposal is to present the design, manufacturing, and supply of the PTL system.

3. Pick/Put to Light Module

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with Client's WMS/ERP Systems.

The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keep on altering the PTL configuration to increase the system productivity and give you the extra competitive edge.

The System involves a collection of physical sorting locations representing either a customer order/ address location or any other logical sorting group. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.

It is a paperless technology provide high throughput with single operator with high accuracy.



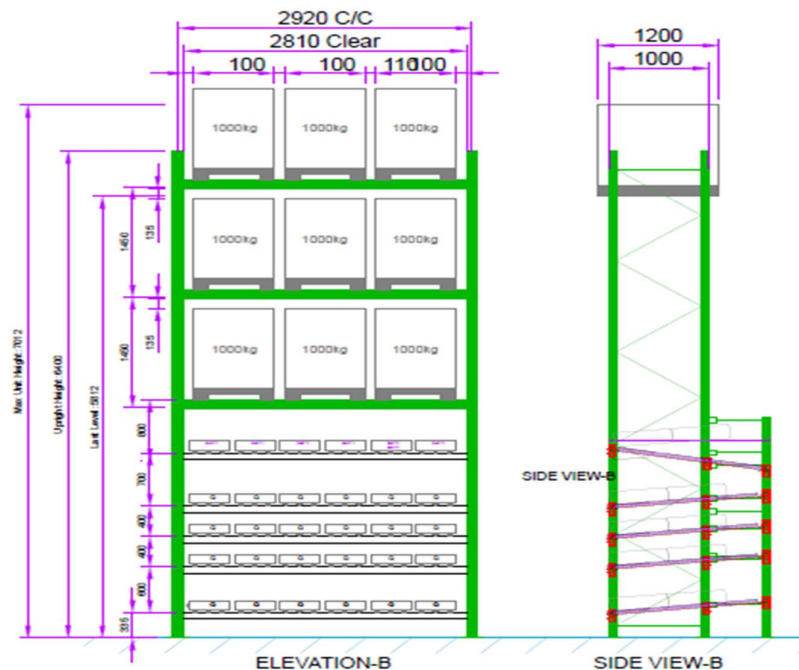
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Dimensions	Mm	Around 165 X 50
PTL location	Nos	As per Layout requirement
LED Display	Type	3 Digit Alpha numeric Rotating Display
LED Size	Mm	~ 25 X 20
LED Indicator	yes/no	YES
Acknowledgment Button	yes/no	YES
Function keys	yes/no	4 Keys
Power Requirements	Type	3 Phase UPS Power
Communication	Type	TCP/IP and Serial
Operating temperature	Deg.	45 Degrees
Software Interface	Type	Over APIs or other methods
Type of mount	Type	Clip On Type on Rails

4. Sample Rack Layout

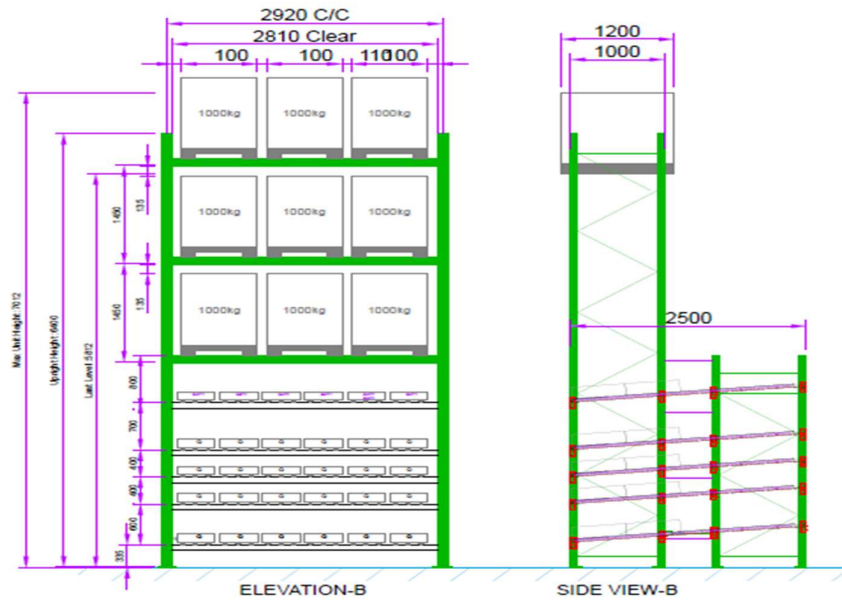
4.1 Type-1



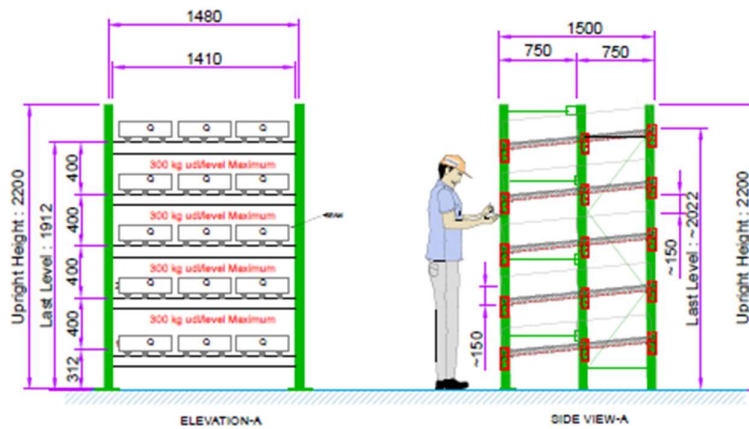
Note: Above image is for reference purposes only, it may change depending on the configuration of the rack.



4.2 Type-2



4.3 Type-3

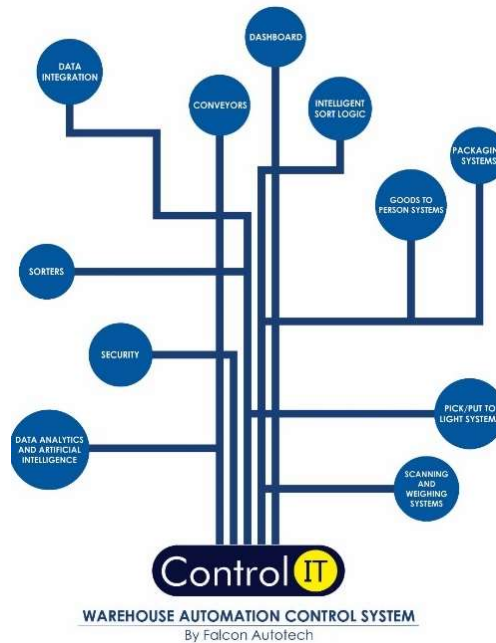


5. Handheld Barcode Scanner

Each PTL Frame will be equipped with a Handheld wireless 2D Barcode scanner.



6. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

7. Proposed System Technical Details/BOM

7.1 Mechanical equipment

7.1.1 Setup of Type-1 (36 X 30 Lights)

Pos.	Qty.	Description	Value
FM1	1	PTL System	
		PTL Frame	36 Nos
		PTL Modules	1080 Nos
		PTL Control Panel	36 Nos
		Handheld Scanner	36 Nos

7.1.2 Setup of Type-2 (22 X 30 Lights)

Pos.	Qty.	Description	Value
FM1	1	PTL System	
		PTL Frame	22 Nos
		PTL Modules	660 Nos
		PTL Control Panel	22 Nos
		Handheld Scanner	22 Nos

7.1.3 Setup of Type-3 (10 X 30 Lights)

Pos.	Qty.	Description	Value
FM1	1	PTL System	
		PTL Frame	10 Nos
		PTL Modules	300 Nos
		PTL Control Panel	10 Nos
		Handheld Scanner	10 Nos

7.2 Electrical Equipment

Electricals			
FC1	1	Electrical Cabling and Wiring	
		Field Cabling	Included

7.3 Control System

Components			
FC2	1	Controls and IT Infrastructure Controls	Included
		PTL Controls	
		IT	
		Control IT Software Package	

8. Commercial Terms

8.1 Price Sheet

S. No	Component	Set	Price (USD)
1	PTL System	1	\$ 2.57.179,00
2	Pallet Racking & Bin flow racks	1	\$ 1.98.750,00
3	Software Packages	1	\$ 7.692,00
4	Packaging & forwarding	1	\$ 22.796,00
5	Project Management and Engineering Charges	1	\$ 12.160,00
Total (USD)			\$ 4.98.579,00

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Freight charges- In client scope.
- Taxes and duties shall be extra as applicable in client scope.
- Installation & commissioning- Excluded.
- Including Packing charges
- Duty unpaid
- Price is valid for 60 days from the date of proposal.

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

*Material Unloading & Laydown area is in customer scope

9. Dispatch Timelines

Dispatch within 12-14 weeks from the date of PO, DAP & Advance.

10. Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance along with PO.

Stage 2: 50% along with 100% Taxes prior to dispatch.

11. Warranty Period

Falcons offered System comes with a standard warranty of 1 year for complete system against any manufacturing defect. Any damage caused to the system by an external source, abnormal use of equipment or any other factor not attributable to Falcon Autotech shall not be covered under warranty.

Warranty covers the following support:

- Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

*Manpower support required for maintenance of the system is not included in this proposal.

12. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Server PC.
- Steelworks if not specified.
- Manpower support for maintenance.
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives.
- Electrical Power for Installation.
- Network Cabling up to the Main Server Rack.
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply.
- Fire-alarm and fire protection devices.
- Traffic and route markings.
- Laydown Area.
- Ram protection devices.
- Cat walks, bridges, maintenance aisles and platforms.
- Assembly tools like forklifts and hoisting machines.
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply.
- Any Kind of Civil work.
- Any adjustment of the Supplier's scope of supply to local rules and regulations.
- X-Ray machines.
- Roller cages/ Pallets.
- Simulation and 3D animation of the sorter system.
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.